

BENJAMIN ADJEPONG

US Citizen - Eligible for Clearance

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Education

Virginia Polytechnic State Institute & University August 2024 – May 2026
Master of Engineering in Computer Engineering, Software and Machine Intelligence — GPA: 3.85 Blacksburg, VA

Virginia Polytechnic State Institute & University August 2021 – May 2025
Bachelor of Science in Computer Engineering, Machine Learning — GPA: 3.64 Blacksburg, VA

Relevant Coursework: Artificial Intelligence, Brain-Inspired Computer Architecture, Computer Vision, Machine Learning, Trustworthy Machine Learning, Video Game Design, Data Structures & Algorithms

Experience

Microsoft May 2025 – August 2025
Software for Hardware Engineer Intern – Test Automation & Reporting Redmond, WA

- Automated test execution using PowerShell scripts to parse and extract test parameters before triggering test framework
- Identified and integrated diverse data sources into a scalable framework for reporting and continuous monitoring
- Consolidated test data into a centralized Power BI dashboard for streamlined viewing by diverse stakeholders

Microsoft May 2024 – August 2024
Software for Hardware Engineer Intern – Log Based Anomaly Detection Redmond, WA

- Established an AI/ML framework for operation health assessment of devices utilizing log files
- Cleaned and parsed, transforming raw log data into a structured format, enabling thorough log data analysis
- Trained and deployed an unsupervised model to identify anomalous data points in device log dataset

Collins Aerospace June 2023 – August 2023
SEPP Mission Systems Software Engineer Intern – Station Keeping Playbook (SKP) Sterling, VA

- Developed operational testing platform for unmanned aerial systems on ground robots reducing costs by up to 90%
- Tested autonomous capabilities for UAS's by integrating autonomous flight software onto ground robots
- Learned and implemented various python libraries to process GPS & IMU data to command motors based on SKP

Projects

Machine Learning Integrated Biosensor For Disease Detection | PM, Python August 2024 – May 2025

- Led cross-functional team in development of machine learning based biosensor system for SCM detection in cattle
- Facilitated regular team meetings with customers and created agendas & budgets to ensure alignment with goals
- Conducted literature surveys and review research papers to familiarize team with industry standards and best practices
- Designed and implemented machine learning submodule for milk classification using Raman and Impedance spectra data

Facial Attribute Detection | CV, Python September 2024 – December 2024

- Developed a real-time face detection system using CNN to identify key features, such as facial geometry and so on
- Trained model on diverse dataset of facial characteristics to improve detection accuracy and attribute classification
- Evaluated against alternative implementations, uncovering optimization opportunities in performance improvements

Autonomous Tractor | Arduino, Python January 2023 – May 2023

- Designed and implemented a graphical user interface (GUI) with PyGame for real-time tractor control and monitoring
- Integrated Bluetooth Low Energy (BLE) communication to enable wireless data exchange between the GUI and tractor
- Connected and programmed an Arduino Uno with 4+ sensors to enable autonomous navigation and obstacle detection

Skills

Languages: Python, PowerShell, KQL, C/C++, JavaScript, Arduino, R, Verilog/VHDL

Developer Tools: scikit-learn, VS Code, GitHub, Jupyter Notebook, ChatGPT, Azure ML Studio, Azure DevOps

Techincal Tools: Nvidia Jetson Nano, TI MSP432 Launchpad & Boosterpack, DE10-Lite FPGAAD2 Oscilloscope, LTSpice

Extracurricular

National Society of Black Engineers September 2021 – August 2025
Senator Virginia Tech

- Co-led planning and execution of 2 CAMEO Career Fairs, coordination logistics and day-of operations
- Organized and coordinated 5+ social and academic events for the general student body at Virginia Tech